

Marco Sabatini

Professional Summary

Engineering leader with 15+ years of experience in building scalable data platforms, distributed systems, and cloud infrastructure across fintech, cybersecurity, and travel-tech industries. Proven track record in leading engineering teams, designing modern data architectures, and delivering production-grade data platforms for analytics, risk modeling, and data-driven decision making. Strong expertise in data pipelines, big data ecosystems, cloud-native infrastructure, and platform engineering. Passionate about mentoring engineers, building high-performing teams, and aligning technical platforms with business strategy.

Core Competencies

Data Platform Architecture

- Data Lakehouse architectures
- Data ingestion and processing pipelines
- Streaming and event-driven systems
- Data lifecycle management
- Data quality and reliability

Cloud Data Platforms

- AWS & Azure storage solutions
- AWS & Azure RDS databases
- Kubernetes container orchestration
- Infrastructure as Code with Terraform

Data Engineering & Technologies

- Apache Spark for big data processing
- Databricks for data analytics
- SQL and NoSQL databases (Oracle, MySQL, PostgreSQL)
- Elasticsearch and SOLR for search
- Data orchestration and microservices architecture
- Programming: Python, Java, Ruby, TypeScript, Bash

Leadership & Strategy

- Engineering team leadership and management
 - Platform strategy and architecture design
 - Cross-team technical coordination
 - Stakeholder alignment with product and business teams
 - Mentoring and developing engineering talent
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Professional Experience

Associate Director – Software Engineering

UBS – Lugano, Switzerland

November 2024 – Present

Mission:

- I am a member of the AI adoption team, responsible for providing training on agentic software development and identifying use cases for AI in the banking domain.
- Led a team of 15 engineers responsible for migrating financial risk models to cloud-native data platforms supporting enterprise analytics and risk management.
- Led the migration of US mortgage probability of default (PD) and loss given default (LGD) risk modeling workloads from on-premises systems to Azure cloud infrastructure.
- Designed scalable cloud architecture using Azure Kubernetes Service (AKS), Databricks, and cloud storage solutions.
- Implemented Infrastructure as Code (IaC) using Terraform to standardize and automate deployments.
- Improved scalability and reliability of data processing pipelines supporting quantitative risk analysis teams.
- Coached and mentored engineers while promoting engineering best practices across the organization.
- Integration with quantitative analyst about delivering and structuring delivery process for risk models.

Achievements:

- Successfully migrated enterprise-scale mortgage risk models covering a portfolio of approximately 100,000 US mortgages to Azure cloud platforms.
- Delivered a scalable and resilient cloud-native architecture supporting large-scale financial risk analytics workloads.
- Standardized infrastructure provisioning using Terraform, improving deployment consistency and operational efficiency.
- Enhanced performance and reliability of data processing pipelines used by quantitative risk teams.
- Developed engineering talent by coaching four engineers and improving overall team engineering maturity.

Technologies: Kubernetes, Databricks, Python, Terraform, Docker, Oracle Database, Github Copilot, GitLab

Senior Software Architect

Denexus – Lugano, Switzerland

May 2022 – November 2024

Mission:

- Architected and developed a cyber risk intelligence platform for aggregating and enriching large-scale network asset and vulnerability data.
- Designed and implemented cloud-native architecture for cybersecurity data ingestion, enrichment, and indexing.
- Built scalable data processing pipelines using Python and Databricks for intelligence data workflows.
- Developed microservices in Python using FastAPI to enable data discovery and search across industrial networks.
- Collaborated with leadership on platform architecture decisions and roadmap planning.

Achievements:

- Delivered a production cyber risk intelligence platform processing large-scale cybersecurity datasets for industrial environments.
- Enabled risk analysis across approximately 20 sensitive industrial plants in the USA and Europe.

- Built data pipelines capable of processing terabytes of data with optimized model execution completing within minutes.
- Improved organizational capability for cybersecurity intelligence through a scalable, cloud-native platform architecture.

Technologies: AWS (ECS, EC2, S3), Databricks, Python, FastAPI, Terraform, Docker, PostgreSQL, Splunk

Senior Software Engineer

Nozomi Networks – Mendrisio, Switzerland

November 2020 – May 2022

Mission:

- Contributed to the development of an asset-indexing and search platform for industrial sites.
- Implemented Ruby-based APIs for search functionality and Elasticsearch integration.
- Designed indexing schemas to support high-throughput search workloads (hundreds of queries per second).
- Developed Terraform infrastructure as code to provision automated testing and production environments.

Achievements:

- Delivered a scalable search platform supporting high-throughput query processing (hundreds of searches per second).
- Enabled production deployment of an asset-indexing system used by approximately 100 customers.
- Improved system reliability and deployment efficiency through infrastructure automation using Terraform.
- Designed search indexing structures that supported performant and scalable Elasticsearch-based queries.

Senior Software Engineer

lastminute.com – Chiasso, Switzerland

November 2015 – November 2020

Mission:

- Contributed to the flight pricing domain, defining requirements and implementing production features in Kotlin for a high-throughput fare pricing and markup platform.
- Owned production monitoring, incident response, and operational maintenance for business-critical services.
- Led technical recruiting, candidate evaluation, onboarding, and developer training initiatives.
- Designed and delivered a two-week developer bootcamp covering Test-Driven Development (TDD), Agile delivery practices, code quality, and release management.
- Served as area architect for hotel search, defining technical roadmaps, coordinating delivery across three engineering teams, and driving engineering excellence through mentoring, pairing, and code reviews.

Achievements:

- Delivered production features for a large-scale flight pricing engine capable of processing and marking up thousands of fares per second across multiple carriers.
- Ensured reliability and operational stability of critical pricing services through ownership of monitoring, incident response, and production support.
- Established and ran a structured onboarding and training program that accelerated developer productivity and adoption of engineering best practices.
- Led the architecture and delivery of the hotel search domain across three teams, improving technical consistency, delivery quality, and team effectiveness.

- Mentored engineers through hands-on coaching, code reviews, and pairing sessions, contributing to stronger engineering standards and performance.

Earlier Experience

Solution Architect Consultant

Wedjaa – Rome, Italy

2011 – 2015

Mission:

- Worked as a consultant across multiple enterprise projects, leading client engagement, technical assessments, solution architecture, and software delivery.
- Defined solution architectures and developed software for customer-facing and internal platforms.
- Built and shaped internal and external development teams to support project execution.

Achievements:

- Delivered a Java-based ticketing platform for an Italian cinema chain with approximately 30 locations and hundreds of screens, supporting peak holiday demand of around 20,000 daily bookings.
- Implemented an online medical reporting platform for the Lazio region, integrating Alfresco and the open-source eHealth Integration Framework (IPF) for patient data management.
- Delivered a project reporting and accounting platform for an international telecommunications company, including requirements management and Jira integration.
- Successfully led multiple projects from technical assessment through production delivery while coordinating stakeholders and development teams.

System Analyst

IDS – Rome, Italy

2009 – 2011

Mission:

- Contributed to the development of two aerospace systems for flight procedure design and obstacle management in controlled airspace.
- Participated in requirements gathering, feature implementation, and production releases.
- Developed the majority of the backend in Java and implemented frontend components using JavaScript and HTML.

Achievements:

- Delivered core backend functionality for two mission-critical aerospace applications.
- Supported successful production deployments used by approximately ten national and international aviation and airport authority organizations.
- Contributed to the end-to-end delivery process, from requirements definition through release to production.

Software Developer

IBM – Rome, Italy

2008 – 2009

Mission:

- Collaborated with a senior engineer to define and implement features for replacing a proprietary search engine within a publishing lifecycle system serving international magazines.
- Developed a Java-based search engine using Apache Lucene for information retrieval.

- Implemented batch and real-time indexing capabilities for millions of documents per client using Java and socket-based integration with the publishing platform.

Achievements:

- Helped replace a proprietary search engine with a custom Apache Lucene-based solution.
 - Delivered a search platform supporting millions of documents per client in batch and real-time modes.
 - Contributed to a successful production deployment adopted by approximately 30 customers.
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Education

Master of Science in Computer Science

University of Rome Tor Vergata, Italy

Thesis: Integration of Legacy Systems Using SOA Principles

Grade: 107/110

Professional Development

- Advanced Test Driven Development – J.B. Rainsberger
- Surviving Legacy Code – J.B. Rainsberger
- Applied Functional Programming – Avanscoperta
- Machine Learning & Predictive Analytics – Simone Scardapane
- Secure Coding – Nozomi Networks